

## NEVERS-FOURCHAMBAULT AP, France

WMO: 072600

Lat: 46.9983N

Lon: 3.1128E

Elev: 177

StdP: 99.22

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -7.1       | -5.0 | -9.7                          | 1.7 | -5.9 | -7.5 | 2.0 | -3.6 | 9.9                   | 9.1  | 8.8 | 8.6  | 1.1                   | 20   | 0.503 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.5                   | 31.9            | 20.8 | 29.7 | 20.2 | 27.8 | 19.4 | 21.9                | 29.4 | 21.0 | 28.3 | 20.1 | 26.5 | 3.2                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.3                            | 14.4 | 25.2 | 18.5 | 13.6 | 24.0 | 17.7 | 13.0 | 22.9 | 64.6          | 29.4 | 61.3 | 28.4 | 58.2 | 26.7 | 25.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 8.0               | 7.0  | 6.2 | -10.3                      | 34.9 | 2.7                | 2.3 | -12.2                                              | 36.5 | -13.7      | 37.9 | -15.2      | 39.2 | -17.1      | 40.8 |
|                   |      |     | -10.6                      | 23.5 | 2.6                | 0.8 | -12.5                                              | 24.0 | -14.0      | 24.5 | -15.5      | 24.9 | -17.4      | 25.5 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 11.2   | 3.8   | 4.3   | 7.4   | 10.1  | 13.8  | 17.7  | 19.4  | 19.0  | 15.2  | 12.1  | 7.1   | 4.3   |
|                                                                        | DBStd       | 6.69   | 4.25  | 4.02  | 3.56  | 3.34  | 3.39  | 3.43  | 3.15  | 3.32  | 3.11  | 3.95  | 3.99  | 4.30  |
|                                                                        | HDD10.0     | 806    | 195   | 161   | 94    | 39    | 6     | 0     | 0     | 2     | 25    | 103   | 179   |       |
|                                                                        | HDD18.3     | 2752   | 451   | 392   | 338   | 248   | 145   | 52    | 25    | 31    | 103   | 195   | 338   | 434   |
|                                                                        | CDD10.0     | 1251   | 2     | 2     | 14    | 42    | 124   | 231   | 291   | 279   | 156   | 90    | 15    | 4     |
|                                                                        | CDD18.3     | 158    | 0     | 0     | 0     | 0     | 5     | 33    | 58    | 52    | 8     | 2     | 0     | 0     |
|                                                                        | CDH23.3     | 2035   | 0     | 0     | 0     | 15    | 98    | 438   | 691   | 647   | 131   | 14    | 0     | 0     |
|                                                                        | CDH26.7     | 684    | 0     | 0     | 0     | 1     | 14    | 143   | 242   | 256   | 27    | 1     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 2.6    | 2.9   | 3.0   | 3.0   | 2.8   | 2.6   | 2.5   | 2.5   | 2.2   | 2.1   | 2.4   | 2.6   | 2.9   |
| (15) Precipitation                                                     | PrecAvg     | 798    | 60    | 57    | 50    | 70    | 77    | 64    | 63    | 67    | 65    | 75    | 78    | 70    |
|                                                                        | PrecMax     | 1000   | 137   | 109   | 120   | 207   | 154   | 141   | 136   | 161   | 169   | 125   | 157   | 149   |
|                                                                        | PrecMin     | 581    | 9     | 8     | 10    | 3     | 13    | 13    | 9     | 18    | 22    | 27    | 28    | 15    |
|                                                                        | PrecStd     | 98     | 30    | 29    | 32    | 42    | 34    | 29    | 30    | 36    | 33    | 30    | 30    | 36    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 13.8  | 16.7  | 21.6  | 25.3  | 28.6  | 33.7  | 34.2  | 36.1  | 29.6  | 25.2  | 18.6  | 15.0  |
|                                                                        |             | MCWB   | 10.9  | 10.6  | 12.7  | 16.0  | 19.5  | 21.8  | 20.7  | 21.3  | 19.0  | 17.3  | 14.2  | 11.5  |
|                                                                        | 2%          | DB     | 12.5  | 14.0  | 18.8  | 22.4  | 26.0  | 30.1  | 31.5  | 32.0  | 26.9  | 22.6  | 16.0  | 13.1  |
|                                                                        |             | MCWB   | 10.4  | 9.7   | 11.8  | 14.4  | 17.7  | 20.6  | 20.8  | 20.7  | 18.6  | 17.0  | 12.9  | 11.0  |
|                                                                        | 5%          | DB     | 11.3  | 12.0  | 16.4  | 20.0  | 23.9  | 27.9  | 29.3  | 29.4  | 24.5  | 20.2  | 14.5  | 11.7  |
|                                                                        |             | MCWB   | 9.4   | 9.0   | 10.8  | 13.1  | 16.9  | 19.6  | 20.1  | 20.1  | 17.6  | 15.8  | 12.3  | 10.1  |
|                                                                        | 10%         | DB     | 9.9   | 10.5  | 14.2  | 17.6  | 21.6  | 25.7  | 27.3  | 26.9  | 22.3  | 18.3  | 13.0  | 10.5  |
|                                                                        |             | MCWB   | 8.4   | 8.2   | 9.9   | 12.0  | 15.7  | 18.6  | 19.3  | 19.4  | 16.6  | 14.9  | 11.2  | 9.0   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 12.0  | 11.6  | 13.9  | 16.9  | 20.4  | 23.1  | 22.8  | 22.8  | 21.0  | 18.6  | 15.1  | 12.5  |
|                                                                        |             | MCDB   | 12.9  | 14.6  | 19.1  | 23.9  | 26.8  | 31.1  | 30.6  | 30.6  | 26.7  | 23.2  | 17.3  | 13.9  |
|                                                                        | 2%          | WB     | 10.8  | 10.4  | 12.5  | 15.2  | 18.8  | 21.5  | 21.6  | 21.7  | 19.4  | 17.3  | 13.6  | 11.3  |
|                                                                        |             | MCDB   | 12.2  | 12.8  | 16.9  | 21.2  | 24.7  | 28.4  | 29.4  | 29.4  | 24.6  | 21.2  | 15.4  | 12.7  |
|                                                                        | 5%          | WB     | 9.6   | 9.4   | 11.5  | 13.7  | 17.4  | 20.3  | 20.7  | 20.7  | 18.2  | 16.3  | 12.5  | 10.3  |
|                                                                        |             | MCDB   | 11.0  | 11.5  | 15.4  | 18.5  | 22.7  | 26.6  | 28.0  | 28.2  | 23.3  | 19.5  | 14.3  | 11.6  |
|                                                                        | 10%         | WB     | 8.4   | 8.4   | 10.5  | 12.5  | 16.1  | 19.1  | 19.7  | 19.7  | 17.2  | 15.2  | 11.3  | 9.1   |
|                                                                        |             | MCDB   | 9.7   | 10.5  | 13.7  | 16.6  | 20.6  | 24.6  | 26.1  | 25.9  | 21.3  | 17.9  | 12.9  | 10.4  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 6.4   | 8.4   | 11.0  | 12.0  | 11.9  | 12.2  | 12.5  | 12.9  | 12.3  | 10.2  | 7.6   | 6.5   |
|                                                                        |             | MCDBR  | 7.4   | 11.5  | 15.9  | 17.3  | 16.2  | 16.7  | 16.6  | 17.8  | 16.7  | 12.7  | 9.2   | 7.4   |
|                                                                        |             | MCWBR  | 5.7   | 8.0   | 9.7   | 9.9   | 9.0   | 8.0   | 7.3   | 7.8   | 9.0   | 7.7   | 6.5   | 5.7   |
|                                                                        | 5% WB       | MCDBR  | 6.7   | 9.8   | 13.1  | 15.3  | 14.6  | 14.9  | 14.8  | 15.6  | 14.1  | 11.5  | 8.4   | 6.7   |
|                                                                        |             | MCWBR  | 5.5   | 7.2   | 8.4   | 9.1   | 8.5   | 7.5   | 6.8   | 7.3   | 8.0   | 7.3   | 6.2   | 5.5   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.315 | 0.335 | 0.363 | 0.396 | 0.400 | 0.410 | 0.405 | 0.398 | 0.367 | 0.358 | 0.328 | 0.311 |
|                                                                        | taud        |        | 2.455 | 2.400 | 2.329 | 2.236 | 2.272 | 2.283 | 2.295 | 2.329 | 2.393 | 2.420 | 2.455 | 2.447 |
|                                                                        | Ebn at Noon |        | 775   | 830   | 856   | 857   | 865   | 856   | 855   | 846   | 842   | 790   | 748   | 731   |
|                                                                        | Edh at Noon |        | 67    | 87    | 107   | 129   | 129   | 129   | 126   | 117   | 100   | 84    | 67    | 61    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.06  | 1.88  | 3.14  | 4.40  | 5.17  | 5.95  | 5.75  | 5.03  | 3.88  | 2.39  | 1.28  | 0.90  |
|                                                                        | RadStd      |        | 0.14  | 0.28  | 0.38  | 0.54  | 0.50  | 0.55  | 0.43  | 0.45  | 0.30  | 0.18  | 0.14  | 0.15  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | +0.89   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (25 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page